

A. Appendix

A.1. Questions Energy Systems

Review what you learned by answering following questions.

Global Energy Resources

1. The energy flow from mining to consumption is categorized in five stages. List these stages.
2. Give an example for a Primary Energy / Secondary Energy.
3. When talking about energy sources their availability is split into Reserve and Resource. Define the meaning of these two terms Reserve and Resource.
4. Name the three mostly used primary energy sources.
5. Name three renewable energy sources.
6. Name three non-renewable energy sources
7. What does the abbreviation OPEC stands for?
8. Why does the shale oil production received a boost in recent years?

Energy Transformation

9. Name and describe the levels of a traditional hierarchical power system structure.
10. Why does the hierarchical power system has several voltage levels?
11. What is a typical voltages for LV, MV, HV voltage levels?
12. What are the abbreviations DSO and TSO standing for and give one example each?

Electrical Energy Production

13. Describe the terms base load, mid-merit load and peak load.
14. What are typical base load power plants and what is their common characteristic?
15. What is a typical mid-merit power plant?
16. What are typical peak power plants and what is their common characteristic?
17. Describe the term spinning reserve and what is it used for?
18. What does the Wilians line describes?
19. Describe the three levels of the electrical energy market.
20. Which levels of the electrical energy markets are partly regulated (natural monopoly) and which are open for competition?
21. Worldwide we see the trend of unbundling former monopolistic energy companies. Generation becomes separated from transmission and distribution. What is the motivation, driving force behind this development?
22. What are the biggest differences between regular economic market and the electricity market?
23. Because the electrical energy market has some limitations compared to a regular economic market. Therefore the transmission and distribution grid is still regulated. Who is responsible for the regulation in Austria?
24. Why is the tradable product Electrical Energy time based?
25. Why is a central coordination of the electrical infrastructure needed?
26. What are the tasks of a control area manager?
27. Name the control area managers for Austria, Germany and Switzerland.
28. What voltage levels has the Austrian high voltage grid?

Power Generation Costs

29. What are CAPEX and what are they used for?
30. Give examples for CAPEX's.
31. What are OPEX and what are they used for?
32. What is the effective operating time of a power plant?
33. How is the Utilization (dt. Auslastung) of a power plant defined?
34. How is the power generation costs ϵ defined?

Generation System Planning

35. What is the accumulated daily load curve used for?

Energy Trading

- 36. For which primary energy sources are markets existing?
- 37. For which primary energy sources are no markets existing?
- 38. What is traded on the Emission rights market?
- 39. What markets are available for electrical power?
- 40. How can the product *electrical energy* be classified?
- 41. Product characteristics like delivery period and profile are defining the marketplace where the product is being traded. Name two market places and the typical delivery period, delivery profile and trading period of the products traded there.
- 42. Why do have utilities have their own optimal portfolio of electrical energy products?
- 43. What are the two basic ways business transactions are happening?
- 44. Which type of transaction is using an open ordering book and how does it works?
- 45. Which markets are usually doing continuous trading?
- 46. Which type of transaction is using a closed ordering book and how does it works?
- 47. Which market is usually doing auctions?
- 48. What is a Market Clearing Price?
- 49. How are business transactions being fulfilled after the product electrical energy has been delivered?
- 50. What does a physical fulfillment of a business transaction means and for what electricity products is it usually used?
- 51. What does Cash-settlement of a business transaction means
- 52. What does a Cascaded business transaction means and for what electricity products is it usually used?
- 53. How are Options, traded on stock markets, fulfilled?
- 54. What does the abbreviation OTC trading stays for and what does it means?
- 55. What is the role of a broker within an OTC business?
- 56. Name three power exchange trading places in Europe, what products are traded there and which areas are covered.

57. What is the task of the European Commodity Clearing (ECC), the central counter part at EPEX?
58. Which is the most common way to buy/sell electrical energy?
59. How does the Day Ahead Trading system works?
60. What is the main decision factor when a utility has to decide to produce electrical energy by them self or buying it from the market in order to satisfy their obligations.
61. What does the Merit order curve shows?
62. What is a block order?
63. What does the abbreviation Phelix-Index stays for?
64. How does the Intraday Trading system works?
65. What is the Futures Trading system and what is it used for?
66. Who is in charge for trading transmission line capacities?
67. What is a balance group used for and who can participate in such a group?
68. Who is the central clearing and settlement agent (CSA) (dt. Bilanzgruppenkoordinator) in Austria?
69. Give three examples for balance responsible parties (dt. Bilanzgruppenverantwortliche) registered in Austria?

Thermal Plants

70. Name for fuel types used in thermal power plants.
71. What is a cycle process (dt. Kreiprozess)?
72. Name to typically used working materials in cycle processes?
73. Describe the four stages of the Carnot cycle.
74. Draw a typical p-v diagram of an Carnot cycle for an ideal gas.
75. Draw a typical T-s diagram of an Carnot cycle for an ideal gas.
76. Name the typical components of a power plant in which the Rankine cycle is realized.
77. Describe the four stages of the Rankine cycle.
78. Draw a typical T-s diagram of an ideal Rankine cycle.
79. What is the difference between an ideal and an real Rankine process.
80. What is the dryness fraction x (dt. Dampfgehalt) used for?
81. What are the two extreme points of the dryness fraction x describing? ($x=0$, $x=1$)

82. What are the three aggregation states of the working medium steam?
83. Name two application areas of gas turbines.
84. What is a typical start-up time of a gas turbine?
85. Name the typical components of a gas turbine where the Bryton cycle is realized.
86. Describe the four stages of the Brayton cycle.
87. Draw a typical T-s diagram of an ideal Brayton cycle.
88. Name the components of a coal fired power plant.
89. Describe a typical steps of a fixed bed or grate firing process.
90. What are the advantages / disadvantages of a grate firing system with a feed stoker?
91. What is a *Bubbling fluidized bed*?
92. What is a *Circulating fluidized bed* (CFB)?
93. What are the advantages / disadvantages of dust firing?
94. Name three ways a coal burner can be installed in a boiler.
95. What is the difference between primary and secondary air used by an coal burner?
96. Name two methods on how the initial ignition of a coal burner is done.
97. What are the two basic different designs of a steam generator?
98. What does the parameter *circulation ratio* describes within a steam generator?
99. What are condenser used for in a thermal power plant?
100. How can condenser be classified?
101. Explain the working principle of Electrostatic precipitator (ESP).
102. Describe how a *wet scrubber* is used for flue gas desulphurization.
103. Name two methods for NO_x emission control.
104. Name the components of a steam turbine.
105. What is a *turbine stage*?
106. Describe the Reaction Principle of a steam turbine.
107. Describe the Impulse Principle of a steam turbine.
108. What is the working principle of combined-cycle gas-turbine (CCGT) plants?
109. What are the feature of a CCGT?
110. Where does the heat in a nuclear power plant comes from?
111. What does the term *moderator* means in a nuclear power plant?
112. What do the terms *light water* and *heavy water* means in a nuclear power plant?

- 113. Describe *boiling water reactor (BWR)* and *Pressurized Water Reactor (PWR)*.
- 114. Name the two main tasks of a thermal power plant controller.
- 115. Describe the control strategy *Boiler following*.
- 116. Describe the control strategy *Turbine following*.

Renewable Plants

- 117. How can hydro power plants be categorized?
- 118. What are the advantages of using a diversion canal for river power plants?
- 119. Why is the power house placed on the outer river bank on a bending river?
- 120. What is the commonly used turbine for river power plants, and why?
- 121. What load type (base, medium, peak) is usually covered by river power plants?
- 122. What is a commonly used turbine used for storage power plants, and why?
- 123. What are cascaded river power plants and what possible operation modes are used to operate them?
- 124. What is the difference between the tipping operation and swell operation of cascaded river power plants?
- 125. Describe a typical cyclic operation (dt. Wälzbetrieb) of a pump storage power plant.
- 126. Name at least 3 objectives of a pump storage power plant.
- 127. How are *combined* pump storage power plants defined?
- 128. What is a *tandem set* (dt. Tandemsatz) in a pump storage power plant?
- 129. Describe what the operation mode *hydraulic short circuit* means and what it is used for.
- 130. What is a typical value for a pump storage efficiency?
- 131. Name three types of oceanic power plants.
- 132. Name three characteristics of water.
- 133. What are the two parameters the power output of a hydro power plant depends on the most?
- 134. Describe the phenomenon cavitation erosion.
- 135. Name the three most important hydro turbines.
- 136. Describe how a Francis turbine works.
- 137. Describe how a Pelton turbine works.

138. Describe how a Kaplan turbine works.
139. Describe how photovoltaic conversion works.
140. Name the components of a wind turbine?
141. What makes the blades of a wind turbine move? Describe the physical principle.
142. Name three typically used generators in wind turbines?
143. Describe the working principle of an doubly fed induction generator (dt. Doppel-Gespeisten Asynchongenerator).
144. What are the so called Grid Codes?
145. Name two methods which are used to limit power capture if the wind are getting to strong.
146. A traditional way of turbine control is the so called Danish Concept. What does it consists of?
147. What is the advantage of the variable speed control concept?
148. Name the two ways biomass can be used to produce electricity.

Power System Control

149. What are the tasks of a control area manager (dt. Regelzonenführer)?
150. Describe the working principle of an synchronous generator?
151. How is grid frequency related to real power supply?
152. How is grid voltage related to reactive power supply?
153. What are the three-stage procedure consists of, which is used to keep the electrical power balanced within the Continental European Grid?
154. What is the time period for a single incident in the primary control stage?
155. Who participates in the primary control?
156. What is the time period for a single incident in the secondary control stage?
157. Who participates in the secondary control?
158. What is the time period for a single incident in the tertiary control stage?
159. Who participates in the tertiary control?
160. What does the *minutes reserve* means?